

Nitinder Mohan

University of Helsinki, Finland

Ph.D. Scholar in Computer Science

Marie-Curie Fellow

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Education

University of Helsinki

Ph.D. Scholar

Helsinki, Finland

Sept. 2015 - Current

- Supervised by Prof. Dr. Jussi Kangasharju in CoNe research group

Indraprastha Institute of Information Technology (IIIT)

Delhi, India

M.Tech. Computer Science and Engineering (Specialization in Mobile Computing) 2013-2015

- CGPA - 8.5/10.0
- Nominated for **Best Thesis Award**

University College of Engineering (UCoE), Punjabi University

Punjab, India

B.Tech. Computer Engineering

2009 - 2013

- CGPA - 7.55/10.0
- Graduated with first division with distinction
- Ranked amongst top 10 students in batch

Skills

Expertise Areas

- Edge clouds, Cloud computing, Internet-of-Things, Multipath TCP, Service Function Chains, Information-Centric Networks, Distributed Systems

Programming Languages

- Python, R, HTML5, C, C++, Java, $\text{\LaTeX}$

Tools and Technologies

- Linux Kernel, MATLAB, ns3, Spark
- Docker, GraphLab, TensorFlow
- Android Programming, Soot Static Analyzer, OpenStack Cloud Services

Research Experience

In descending chronological order

University of Göttingen

Visiting Researcher

Göttingen, Germany

January, 2018 - May, 2018

Supervisor: Prof. Dr. Xiaoming Fu

- Research in optimal placement of edge servers and virtualized service functions
- Funded by EU FP7 project - CleanSky ITN

NEC research labs, Europe

Visiting Researcher

Heidelberg, Germany

November, 2016 - March, 2017

Supervisor: Dr. Fabian Schneider

- Research in deploying network functions in edge clouds
- Funded by EU FP7 project - CleanSky ITN

Indian Institute of Technology (IIT)

Project Scientist

Delhi, India

June, 2015 - August, 2015

Supervisor: Dr. Vinay Ribeiro

- Research & Development of project "Structured sharing of resources in a Device-to-Device (D2D) Network"
- Funded by Intel Labs, USA

Indraprastha Institute of Information Technology (IIIT)

M.Tech. (with Thesis)

Delhi, India

Aug. 2013 - May, 2015

Supervisor: Dr. Pushpendra Singh

- Research & Development of a system which checkpoints applications in CCN and provides an efficient sender-optimized method of communication in distributed applications
- Awarded GATE scholarship throughout study by Government of India

Central Scientific Instruments Organization (CSIO) - CSIR Labs

Research Intern

Chandigarh, India

Jan. 2013 - June 2013

Supervisor: Dr. Sudesh K. Mittal

- Research & Development of a prediction model which will generate an early warning for an impending landslide using Image Processing and Interpolation.
- In collaboration with Indian Space Research Organization (ISRO)

Teaching Experience

Distributed Data and Infrastructure

Teaching Assistant (T.A.)

University of Helsinki, Finland

Oct., 2017 - Dec., 2017

Instructor: Prof. Dr. Jussi Kangasharju

- Taught concepts related to Distributed Computing, Machine Learning, Edge Computing and Big Data

- Analyzing real-world open-source data utilizing state-of-the-art tools such as Spark, TensorFlow etc.
- Class strength - 40+ students

Distributed Systems

Teaching Assistant (T.A.)

University of Helsinki, Finland

Sept., 2016 - Dec., 2016

Instructor: Prof. Dr. Jussi Kangasharju

- Taught concepts related to distributed systems, networking protocols and Big Data
- Developing and grading programming assignments to test network protocols and Big Data concepts using python and Spark
- Class strength - 60+ students

Cellular Data Networks

Teaching Assistant (T.A.)

IIIT Delhi, India

Aug., 2013 - Dec., 2013

Instructor: Geeta Tripathi

- Taught concepts related to data communications and advanced level cellular communication such as 3G, 4G, WiMax and WiFi
- Class strength - 90+ students

Positions of Responsibility

Jury member for 1st Edge hackathon organized in Helsinki November 2017
Organizer of 3rd CleanSky conference in Helsinki March 2017
Co-organizer of 2nd CleanSky summer school in Helsinki May 2016
Web-admin of 2nd CleanSky summer school website May 2016
Student Representative of M.Tech. 2013 batch Aug. 2013 - May, 2015
Disciplinary Action Committee (DAC) representative of Post-graduates Aug. 2013 - May, 2015
Design Team Head of IIITD Research Showcase event March, 2015
Member of “Byld”-IIIT Delhi coding club Jan., 2014 - Dec., 2014
Member IIIT Delhi Boys Basketball Team Aug. 2013 - May, 2015
Head of Farewell Committee - M.Tech. 2013 batch April, 2014
Captain of Punjabi University’s Basketball Team 2010
Student member of Indian Society of Technical Education (ISTE) 2011 - present
Student member of Association of Computing Machinery (ACM) 2013 - present

Awards and Achievements

1. Awarded EU FP7 **Marie Curie Initial Training Network (ITN)** funding to support Ph.D. studies for three years
2. Invited to attend “**South Asia Workshop On Research Frontiers In Computing**” held at **National University of Singapore (NUS), Singapore** in May, 2014
3. Graduated with B.Tech. with 1st **Division with Distinction** (Rank amongst top 10 students) in May 2013
4. Awarded **Certificate of Merit** for being **All-India Topper in Mathematics in All-India Secondary School Examination** held by Central Board of Secondary Education (CBSE) in 2007

5. Awarded **GATE scholarship** by Govt. of India – Aug, 2013 – May, 2015
6. **Recognition Award** by Director, IIIT Delhi for various duties undertaken as a **Student Council member** and for the assistance provided by designing various **Institute softwares**

Research Projects Involvement

Adaptive and Intelligent Data (AIDA) <i>Collaborative project between University of Helsinki and Aalto University</i> PI: Prof. Dr. Jussi Kangasharju and Prof. Aristides Gionis	Academy of Finland 2018 - 2021
Bright Clouds Dark Clouds (BCDC) <i>Collaborative project digitalizing utilization of renewable energy via Cloud</i> PI: Prof. Dr. Rauli Svento (University of Oulu, Finland) www.bcdcenergia.fi/en/project/	Academy of Finland 2016 - 2020
CleanSky Initial Training Network <i>Collaborative project involving five universities and four industry partners</i> PI: Prof. Dr. Xiaoming Fu (University of Göttingen, Germany) http://www.cleansky-itn.org/	European Union 2014 - 2018

List of Publications (Google Scholar profile)

I. Conference Publications

- [Under Review] **DeCloud: Truthful Decentralized Double Auction**; Aleksandr Zavodovski, Suzan Bayhan, **Nitinder Mohan**, Pengyuan Zhou and Jussi Kangasharju, *IEEE International Conference on Computer Communications, INFOCOM 2019*
- [Under Review] **Practical Discovery of Edge Nodes in the Wild**; Aleksandr Zavodovski, **Nitinder Mohan**, Walter Wong, Suzan Bayhan and Jussi Kangasharju, *14th International Conference on emerging Networking EXperiments and Technologies, CoNEXT 2018*
- [Under Review] **ICON: Intelligent Container Overlays**; Aleksandr Zavodovski, **Nitinder Mohan**, Suzan Bayhan, Walter Wong and Jussi Kangasharju, *17th ACM Workshop on Hot Topics in Networks, HOTNETS 2018*
- Anveshak: Placing Edge Servers In The Wild**; **Nitinder Mohan**, Aleksandr Zavodovski, Pengyuan Zhou and Jussi Kangasharju, *ACM SIGCOMM Workshop on Mobile Edge Communications (MECOMM), SIGCOMM 2018*
DOI: 10.1145/3229556.3229560
- QAware: A Cross-Layer Approach to MPTCP Scheduling**; Tanya Shreedhar, **Nitinder Mohan**, Sanjit K. Kaul and Jussi Kangasharju, *IFIP Networking Conference, IFIP Networking 2018*
Online: arxiv.org/abs/1808.04390
- MUTE: Multi-Tier Edge networks**; Alessio Silvestro*, **Nitinder Mohan***, Jussi Kangasharju, Fabian Schneider and Xiaoming Fu, *5th Workshop on CrossCloud Infrastructures & Platforms (CrossCloud'18), EuroSys 2018* [*joint first authors]
DOI: 10.1145/3195870.3195871

Managing Data in Computational Edge Clouds; Nitinder Mohan, Pengyuan Zhou, Keerthana Govindaraj, and Jussi Kangasharju, *ACM SIGCOMM Workshop on Mobile Edge Communications (MECOMM)*, SIGCOMM 2017

DOI: 10.1145/3098208.3098212

Edge-Fog Cloud: A Distributed Cloud for Internet of Things Computations; Nitinder Mohan and Jussi Kangasharju, *IEEE International Conference on Cloudification of the Internet of Things (CIoT)*, 2016

DOI: 10.1109/CIOT.2016.7872914

II. Journal Publications

Placing it right!: optimizing energy, processing, and transport in Edge-Fog clouds; Nitinder Mohan and Jussi Kangasharju, *International Journal on Annals of Telecommunication*, Springer, 2018

DOI: 10.1007/s12243-018-0649-0

III. Posters and Demos

Redesigning MPTCP for Edge Clouds; Nitinder Mohan, Tanya Shreedhar, Aleksandr Zavodovski, Otto Waltari, Jussi Kangasharju and Sanjit K. Kaul, *24th International Conference on Mobile Computing and Networking*, MobiCom 2018

Grouping Computational Data in Resource Caches of Edge-Fog Cloud; Nitinder Mohan, Pengyuan Zhou, Keerthana Govindaraj, and Jussi Kangasharju, *4th Workshop on CrossCloud Infrastructures & Platforms*, EuroSys Conference 2017

DOI: 10.1145/3069383.3069391

CCNCheck: Enabling Efficient Checkpointed Applications in Content Centric Networks; Nitinder Mohan and Pushpendra Singh, *CCNxCon*, 2015

Online: <https://arxiv.org/abs/1506.00391>

IV. Thesis

Enabling Sender-initiated Distributed Applications and Checkpointing in Content Centric Networks; Nitinder Mohan and Pushpendra Singh, *Masters Thesis, IIIT Delhi, India*, 2015

Online: <https://repository.iiitd.edu.in/jspui/bitstream/handle/123456789/336/MT13069.pdf>

Conference Visits

ACM SIGCOMM, 2018	Budapest, Hungary
IFIP Networking, 2018	Zürich, Switzerland
EuroSys, 2018	Porto, Portugal
International Conference on Networked Systems (NetSys), 2017	Göttingen, Germany
EuroSys, 2017	Belgrade, Serbia
IEEE International Conference on Cloudification of IoT (CIoT), 2016	Paris, France
3 rd Helsinki-HKUST-Tsinghua workshop on Edge computing, 2016	Helsinki, Finland
Aware-to-Aware machine communication, GI-Dagstuhl, 2016	Dagstuhl, Germany
ACM SIGCOMM, 2016	Florianopolis, Brazil

International/ National level Examinations Qualified

1. **Graduate Aptitude Test Examination (GATE) -2014** (Computer Science), conducted by IIT Kharagpur **All-India Percentile 98.57** (~2,000,000 candidates appeared)
2. **Test of English as Foreign Language (TOEFL)** (valid till 2017) **115/120**
3. **Graduate Records Examination (GRE)** (valid till 2019) **313/340 AW - 3.5/6.0**